- Ass 4 due 30 Oct @ 4pm

Recap:

(1) For quantitative data coming from a one-way layout:

$$Y_{ji} = \mu_j + \varepsilon_{ji}, \quad \text{where } j = 1, 2, \ldots, J \text{ indexes the group}, \quad i = 1, 2, \ldots, n_j \text{ indexes the observations}$$
$$\& \ \varepsilon_{ji} \overset{iid}{\sim} N^*(0, \sigma^2)$$

the GLRT for testing $H_0: \mu_1 = \mu_2 = \ldots = \mu_J$ versus H_1: not all means are equal

reduces to large values of the F-statistic:

$$\frac{SS_{groups}/J-1}{SS_{resid}/N-J} \sim F_{J-1, N-J}$$

The F-statistic is also known as a signal-to-noise ratio.

We can summarise all our steps from last lecture into an ANOVA table:

Source	df	SS	MS	F	Pr(F)
Groups	$J-1$	$\sum_{j=1}^{J} n_j(\bar{Y}_{j\cdot} - \bar{Y}_{\cdot\cdot})^2$	$SS_{groups}/J-1$	MS_{groups}/MS_{resids}	$P(F_{J-1, N-J} \geq F)$
Resids	$N-J$	$\sum_{j=1}^{J}\sum_{i=1}^{n_j}(Y_{ji} - \bar{Y}_{j\cdot})^2$	$SS_{resids}/N-J$		
Total	$N-1$				

In practice, we would never do all these calculations by hand. Instead, we can use R, Matlab, Python, etc... to help us.

Example 8.7 (Movie genres & Rotten tomato scores)

Data from Rotten Tomatoes were collected on movie ratings of Hollywood releases in 2012.

The mean & sds of movies scores, classified by genre, are tabulated below:

Genre	mean	sd	n
Action	44.97	25.10	31
Comedy	49.78	26.35	27
Horror	41.29	27.04	17
Romance	47.20	31.57	10
Animation	60.58	26.73	12
Thriller	62.38	28.66	13
Overall			110

Q: Are there any differences in average movie ratings across genres?

Overall mean rating of movies in 2012:

$$\bar{y}_{..} = \frac{31 \times 44.97 + 27 \times 49.78 + \ldots + 13 \times 62.38}{110}$$

$$= 49.55$$

What is the overall variance of movie ratings?

$$s^2_{pooled} = \frac{30 \times (25.10)^2 + 26 \times (26.35)^2 + \ldots + 12 \times (28.66)^2}{104}$$

$$= 724.4 = (26.91)^2$$

$$= \frac{75337.4}{104}$$ — SSresid / N-J

Finally, $SS_{groups} = 31 \times (44.97 - 49.55)^2$

$+ 27 \times (49.78 - 49.55)^2$

$+ 17 \times (41.29 - 49.55)^2$

$+ 13 \times (62.38 - 49.55)^2$

$= 5466.63$

So now we can construct our ANOVA table:

Source	df	SS	MS	F	$Pr(F_{5,104} \geq 1.509)$
Genre	5	5,466.63	1093.3	1.509	0.193
Resid	104	75,337.4	724.4		
Total	109				

Conclusion: NO evidence that the mean movie ratings are different across the 6 genres.

BUT(!) A movie studio ~~exec~~ executive noticed that "thrillers" had a much higher rating than "horror" movies. A two-sample t-test carried out on the two groups gives:

$$T = \frac{\bar{Y}_{6.} - \bar{Y}_{3.}}{S_{pooled}\sqrt{1/13 + 1/17}} = \frac{62.38 - 41.29}{26.91\sqrt{1/13 + 1/17}} = \frac{21.09}{9.92}$$

$= 2.13$ on $df = 104$

So p-value here is $P(T_{104} \geq 2.13) = 0.018 = 1.8\%$ (!

Based on this post-hoc analysis, the movie executive decides to rebrand his upcoming "horror" film to - "thriller" in hopes of getting higher ratings. Thoughts?

How do we reconcile this apparent contradiction?

There will always be a "best" group & a "worst" group in any multi-group experiment. In this year, it was "thriller" vs "horror", but in another year it could have been "comedy" versus "action", etc...

If we choose to test "best" vs "worst" AFTER looking at our data, then we have to take into account the fact that we could have chosen any of the $\binom{6}{2} = 15$ pairwise comparisons

A quick-&-dirty way of correcting for multiplicity is to use Bonferroni's correction: multiply the p-value by the total number of possible comparisons we could have made.
In this example, the corrected p-value would be $0.018 \times 15 = 0.27$, which is now consistent with our overall ANOVA analysis (0.19)

In general, we should only look at pairwise comparisons after the overall ANOVA returns a significant result. This protects us from inflated Type I errors.

Contrasts: comparing "best" vs. "worst", or indeed any pairwise comparison, is an example of a "contrast". In general, a contrast is any linear combination

$$\sum_{j=1}^{J} a_j \mu_j \quad \text{of the group means}$$

$$\text{s.t} \quad \sum_{j=1}^{J} a_j = 0$$

For example, comparing "thriller" (group 6) versus "horror" (group 3), this

$$\mu_6 - \mu_3 = \sum_{j=1}^{6} a_j \mu_j$$

where $\underset{\sim}{a} = (0, 0, -1, 0, 0, +1)$

This is indeed an example of a contrast!

Contrasts are useful when we want to consider particular comparisons of our group means (not just pairwise). Two popular sets of contrasts arise from the so-called "sum parametrization/constraint" & "contrast constraint" of the ANOVA set-up.

"Sum-parametrization/~~cont~~ constraint":

The one-way ANOVA can be rewritten as

$$Y_{ji} = \underbrace{\mu}_{1 \text{ parameter}} + \underbrace{\alpha_j}_{J \text{ parameters}} + \varepsilon_{ji}, \quad \begin{matrix} j = 1, \ldots, J \\ i = 1, \ldots, n_j \end{matrix}$$

$$\varepsilon_{ji} \overset{iid}{\sim} N^*(0, \sigma^2)$$

where, for identifiability, we can set $\sum_{j=1}^{J} \alpha_j = 0$

In this case, μ is interpreted as the overall mean of all groups,

$$\mu = \frac{\mu_1 + \mu_2 + \ldots + \mu_J}{J}$$

& each α_j is interpreted as the difference between group j & the overall mean μ:

$$\alpha_j = \mu_j - \mu$$

In the sum parametrisation, each α_j is itself a contrast:

Eg

$$\begin{aligned}
\alpha_1 &= \mu_1 - \mu \\
&= \mu_1 - \frac{\mu_1 + \mu_2 + \ldots + \mu_J}{J} \\
&= \left(1 - \tfrac{1}{J}\right)\mu_1 - \tfrac{1}{J}\mu_2 - \tfrac{1}{J}\mu_3 - \ldots - \tfrac{1}{J}\mu_J \\
&= \sum_{j=1}^{J} a_j \mu_j
\end{aligned}$$

where

$$\begin{aligned}
\underset{\sim}{a} &= \left(1 - \tfrac{1}{J}, -\tfrac{1}{J}, -\tfrac{1}{J}, \ldots, -\tfrac{1}{J}\right) \\
&= \left(\frac{J-1}{J}, -\tfrac{1}{J} - \tfrac{1}{J}, \ldots, -\tfrac{1}{J}\right)
\end{aligned}$$

& $\sum_{j=1}^{J} a_j = 0$!

When is the sum-parametrisation useful?

- When all J groups in the experiment are variations of the same treatment.

OR - if the J groups in your study exhaust all possible subgroups of that factor.